

ADA Global Academy - MicroMasters in Data Science, AI & Research Methods

Sprint 01 · Individual Assignment · Due: Friday 20 June 2026, 11:59 PM (WAT)

Child Undernutrition in Nigeria

An Exploratory Analysis of Underweight (WAZ) and Wasting (WHZ) using the NDHS 2024

Dataset	Nigeria DHS 2024 Children's Recode - NGKR8BFL (1).csv
Outcomes	WAZ (underweight) and WHZ (wasting)
Deadline	Friday 20 June 2026, 11:59 PM WAT
Submit	One .R file + one .pdf report, both named Sprint01_FirstnameLastname

1. Background

We spent class looking at stunting (HAZ) - chronic, long-term malnutrition that shows up as a child being too short for their age. This assignment asks you to look at the other two sides of the same problem:

- **WAZ (Weight-for-Age):** Is the child too light for their age? This catches both recent and long-term malnutrition. A WAZ below -2 is called *underweight*.
- **WHZ (Weight-for-Height):** Is the child too thin for how tall they are? This catches recent, sudden malnutrition, for example after an illness or food shortage. A WHZ below -2 is called *wasting*, and it can be life-threatening.

Both are stored in the same CSV file you used in class, but multiplied by 100 (so -152 means -1.52). Values of 9996 or higher are invalid codes, replace them with **NA**, exactly as we did for HAZ.

2. Variables

You are working with 15 variables this time. The table below tells you what each one is and how to bring it into R.

Code	Name	What it is
hw71	WAZ	Weight-for-age z-score
hw72	WHZ	Weight-for-height z-score
hw1	Age (months)	Child's age at measurement
b4	Sex	Child's sex
b5	Child alive	Alive at survey time
m4	Breastfeeding	Breastfeeding status
v024	Region	Geopolitical zone (1-6)
v025	Residence	Urban or rural
v106	Mother's educ.	Mother's highest schooling
v701	Father's educ.	Father's/partner's schooling
v130	Religion	Household religion
v190	Wealth	Household wealth quintile
v136	Household size	Total people in household
v137	Under-5s	Children under 5 in household
v005	Weight	Survey sample weight

A note on **m4** (breastfeeding): this variable has a few unusual codes. Code 93 means the child was never breastfed. Code 94 means the child is still being breastfed at the time of the survey. Code 95 means breastfeeding has stopped. Everything else (0–30) is the duration in months. For this assignment, recode to the three groups above and treat all other values as **NA**.

3. Tasks

3.1 Task A - Data preparation

Load the CSV, select the 15 variables above, and produce a clean dataset. Specifically:

1. Create **waz** and **whz** columns (divide by 100, replace invalid codes with **NA**)
2. Recode all category variables with readable labels using **recode()** or **case_when()**
3. Set ordered factors for wealth (Poorest to Richest) and education (None to Higher)
4. Print: total rows, number of valid WAZ observations, number of valid WHZ observations, and % missing for each

*Start point: copy Step 1.3 from your class script and add the new variables inside the same **mutate()** block.*

3.2 Task B - Descriptive statistics [20 marks]

1. Missing value table - % missing for every variable, sorted highest to lowest
2. Summary statistics for WAZ and WHZ - Mean, Median, SD, Min, Max, Skewness in one table
3. National prevalence table with four rows: Underweight ($WAZ < -2$), Severe underweight ($WAZ < -3$), Wasting ($WHZ < -2$), Severe wasting ($WHZ < -3$)
4. A single comparison table that puts stunting (HAZ, from class), underweight, and wasting side by side, which is most common in Nigeria?
5. Underweight prevalence ($WAZ < -2$) broken down by: region, residence, wealth quintile, mother's education, and religion

Write 2–3 sentences under each table explaining what the numbers mean for child health in Nigeria, not just what they are.

3.3 Task C - Visualisation

Every figure needs a title, a subtitle that mentions “Nigeria NDHS 2024”, axis labels, and a short caption in your report. Save each one with `ggsave()`.

Fig.	Type	What to show
1	Histogram	WAZ distribution. Mark the -2 and -3 cutoffs with vertical lines. Put the underweight % in the subtitle.
2	Histogram	WHZ distribution. Same layout as Fig. 1 but for wasting.
3	Box plot	WAZ by region, sorted from worst to best median. Add the -2 cutoff line.
4	Violin plot	WHZ by wealth quintile, all five groups, -2 cutoff shown.
5	Violin plot	WAZ by breastfeeding status (Never / Still breastfeeding / Stopped). Is there a visible difference?
6	Bar chart	Mean WAZ by religion. Add error bars (use <code>stat_summary</code> with <code>fun.data=mean_se</code>).

3.4 Task D - Correlation and outliers [20 marks]

Correlation: Build a Spearman correlation matrix using WAZ, WHZ, wealth (v190), mother’s education (v106), household size (v136), and number of under-5 children (v137). Plot it as a heatmap (Figure 7) using `corrplot()`. Make a ranked pairs table. In your report, answer: which single variable has the strongest link with WAZ? Is it the same for WHZ?

Outliers (10 marks). Apply the IQR method and the $|z| > 3$ method to both WAZ and WHZ. Produce the summary table and a side-by-side box plot (Figure 8) showing flagged points in orange and the mean as a diamond. Then fill in this table in your report:

Variable	IQR flagged	% flagged	Main region(s)	Decision	Reason
WAZ				Keep / Remove	
WHZ				Keep / Remove	

3.5 Task E - Research questions and conclusion

Three research questions: Based on what you found in Tasks A-D, write three research questions you would want to answer in a future sprint. Each question must name a specific outcome (WAZ or WHZ), a specific subgroup, and be grounded in a pattern you actually saw, say which table or figure showed it.

Example of a weak question: “Does religion affect malnutrition?”

Example of a strong question: “Among children in the poorest wealth quintile, is WHZ significantly lower in Muslim-majority Northern states than in Christian-majority Southern states, after adjusting for household size?”

RQ	Research question	Evidence (table/figure no.)	Suggested method
RQ1			
RQ2			
RQ3			

Conclusion: Include at least three specific numbers. Compare your WAZ and WHZ findings with the HAZ results from class. Name the most at-risk subgroup. Suggest what should be done next analytically.

4. Submission

Submit two files by the deadline:

- `Sprint01.FirstnameLastname.R` - your complete script. It must run from top to bottom without errors when the CSV is in the same folder.
- `Sprint01.FirstnameLastname.pdf` - your written report with all eight figures embedded, all tables, interpretations, and the completed decision and RQ tables.

Before submitting, check: script runs with no errors · all 8 figures saved and in the PDF · every table has an interpretation · decision table filled in · RQ table filled in · conclusion is 150-200 words.

5. Marking rubric

Task	Area	What earns full marks
A	Data prep	All 15 variables loaded, WAZ/WHZ cleaned, factors ordered, summary printed
B	Descriptive stats	All 5 tables with correct cutoffs and 2–3 sentence interpretations
C	Visualisation	All 6 figures with titles, axis labels, captions; saved with <code>ggsave</code>
D1	Correlation	Heatmap and ranked table correct; strongest-predictor question answered
D2	Outliers	Both methods applied to WAZ and WHZ; box plots produced; decision table complete
E1	Research questions	3 specific, grounded, testable questions; each references a figure or table
E2	Conclusion	150-200 words; ≥ 3 numbers; comparison with HAZ; at-risk group named
	Code quality	Commented, readable, runs clean, no stray <code>install.packages()</code> calls

6. A few practical notes

Working directory. The single most common problem is R looking in the wrong folder for your CSV. The fix: go to *Session > Set Working Directory > To Source File Location* at the start of every session.

When something breaks. Read the red error text, R is usually very specific about what went

wrong. Copy it into Google. If you are still stuck after 30 minutes, post in the group chat with the error message and the lines of code around it. Do not just say “it doesn’t work.”

On the new variables. Father’s education (v701) has a code 8 meaning “don’t know”, recode that to NA. Breastfeeding (m4) has codes 93, 94, 95 as explained in Section 2, set anything else to NA. Religion (v130) may have more than three categories in the raw data; collapse anything that is not Christianity or Islam into “Other”.

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